monitoring, predictive analytics and outlier detection, and security threat monitoring and detection.

Elasticsearch focuses on finding a deeper meaning in the data that's been collected. It can aggregate data, perform statistical calculations, and find intelligence within the data. We can create rich visualizations and dashboards and share them with others using Kibana tooling. Elasticsearch can find averages, sums, means, and modes as well as undertaking complex analytics such as bucketing data in histograms and other analytical functions.

Furthermore, Elasticsearch runs supervised and unsupervised machine learning algorithms on our data. Models help to detect anomalies, find outliers, and forecast events. In supervised learning mode, we can provide training sets so the model learns and makes predictions.

Elasticsearch also comes with the capability to observe applications and their health by monitoring performance metrics such as the memory and CPU cycles of the web servers in a network. It lets us sift through millions of web server logs to find or debug application issues. Elasticsearch also invests time and resources in building security solutions: for example, alerting us to security threats, IP filtering, endpoint prevention, and more.

1.4.1 Core areas

Elastic, the company behind Elasticsearch, has been positioning itself predominantly in three core areas: search, observability, and security, as shown in figure 1.2. Let's look at each of these areas in turn.

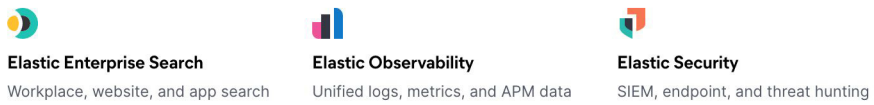


Figure 1.2 Core application areas of Elastic, the company behind Elasticsearch

ELASTIC ENTERPRISE SEARCH

Whether letting users search across varied content providers (like Slack, Confluence, Google Drive, and others) or enabling search capabilities for our applications, apps, and websites, the Elastic Enterprise Search suite helps build models and a customized search engine.

Search can be integrated deep into a multitude of applications in various domains—business, infrastructure, applications, and so on. Users can create a web application backed by Elasticsearch, a mobile app supported by Elasticsearch, or a server-side search service with Elasticsearch as the spine for search capabilities. Later in this book, we work on examples of integrating with Elasticsearch as a search server for applications.